

VALIDATION REPORT

OPERATIONAL QUALIFICATION

CLINICAL STUDY DESIGN MODULE

JR Validated Environment — Clinical Study Design Module v1.0

Field	Value
Document Number	JR-VR-CLIN-001
Title	Validation Report — Operational Qualification, Clinical Study Design Module
Validation Plan	JR-VP-CLIN-001 v1.0
System	JR Validated Environment — Clinical Study Design Module
Module Version	1.0
Document Version	1.0
Execution Date	2026-07-14
OQ Result	PASS — 36 / 36 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-07-14	Joep Rous	Initial OQ execution report. 36 / 36 test cases passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-07-14
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Clinical Study Design Module, version 1.0. It provides objective evidence that all clinical study design scripts perform as specified in the OQ Validation Plan JR-VP-CLIN-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 36 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by `admin_clinical_oq`.

2. Scope

2.1 In Scope

- All 3 R scripts in `repos/clinical/R/` (`jrc_clinical_ss_means`, `jrc_clinical_ss_props`, `jrc_clinical_ss_survival`)
- Two-arm parallel-group sample sizes for superiority, non-inferiority, and equivalence (TOST) frameworks (survival: superiority and non-inferiority only)
- Numeric correctness against independently derived references (Chow, Shao & Wang 2008 ch. 3/4 normal approximations; Schoenfeld 1983 events formula)
- Framework-specific alpha and sidedness reported with the implied z-value
- Allocation ratio, dropout inflation, and sensitivity scenario tables computed in the validated R layer
- Correct rejection of invalid or incomplete designs with informative error messages
- Bypass-protection verification (`RENV_PATHS_ROOT` check)

2.2 Out of Scope

- Infrastructure scripts in `bin/` and `admin/` (covered by JR-IQ-001)
- The Streamlit GUI wizard (`app/clinical_design.py`) — pure orchestration; all figures come from the scripts validated here
- Performance qualification (PQ) with real study protocols
- Validation of the R interpreter
- Statistical validation of the published methods themselves (Chow, Shao & Wang 2008; Schoenfeld 1983 — referenced in JR-VP-CLIN-001)

3. Reference Documents

Document ID	Title	Location
JR-VP-CLIN-001 v1.0	Validation Plan — OQ, Clinical Study Design Module	<code>repos/clinical/docs/clinical_validation_plan.pdf</code>
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite	<code>docs/oq_validation_plan.docx</code>
JR-VP-001 v1.0	Validation Plan — Installation Qualification	<code>docs/</code>
JR-IQ-001	IQ Execution Evidence	<code>docs/IQ_validation_20260311_205146.txt</code>
OQ Evidence	Clinical OQ Execution Evidence (pytest output + header)	<code>clinical_oq_execution_20260714T214209.txt</code>

4. Test Environment

Component	Details
Execution date / time	2026-07-14T21:42:09
Host	Joeps-MacBook-Air.local
Operating System	[not found]
R Version	4.6.0
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/clinical/admin_clinical_oq
Test directory	repos/clinical/oq/
Test data	(none — the scripts are parameter-only)
Execution command	bash repos/clinical/admin_clinical_oq
OQ evidence file	clinical_oq_execution_20260714T214209.txt

5. Test Data

The clinical study design scripts are parameter-only: a sample size is computed before any study data exist, so every test case drives a script through jrrun with command-line flags alone. No CSV fixtures are used. The numeric reference values asserted by the test cases are derived independently of the scripts (documented per test case in the OQ test docstrings and in JR-VP-CLIN-001).

6. Test Results

All 36 test cases were executed on 2026-07-14. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

6.1 Summary

Script	Tests Executed	Result
jrc_clinical_ss_means	14	14/14 PASS
jrc_clinical_ss_props	11	11/11 PASS
jrc_clinical_ss_survival	11	11/11 PASS
TOTAL	36	36/36 PASS

6.2 Results by Test File

Test File	TCs	Result
test_clinical_ss_means.py	TC-CLIN-MEANS-001..014	14/14 PASS
test_clinical_ss_props.py	TC-CLIN-PROPS-001..011	11/11 PASS
test_clinical_ss_survival.py	TC-CLIN-SURV-001..011	11/11 PASS

6.3 jrc_clinical_ss_means — 14/14 PASS

TC ID	Description	Expected	Result
TC-CLIN-MEANS-001	Superiority valid inputs → exit 0, report header	Exit 0, 'Clinical sample size' + 'continuous endpoint'	PASS

TC-CLIN-MEANS-002	Superiority numeric (sd=10, $\delta=5$, power .80, α .05/2)	n treat = 63, n control = 63, n TOTAL = 126	PASS
TC-CLIN-MEANS-003	Sidedness reported with z-value	'2-sided' and z = 1.9600 in output	PASS
TC-CLIN-MEANS-004	Non-inferiority numeric (sd=8, M=4, power .90, α .025/1)	n treat = 85, n control = 85, n TOTAL = 170	PASS
TC-CLIN-MEANS-005	Dropout 0.15 inflation on TC-004 inputs	'ENROLL 100 + 100 = 200' in output	PASS
TC-CLIN-MEANS-006	Equivalence TOST numeric (sd=10, M=5, power .80)	n TOTAL = 138, 'equivalence (TOST)' in output	PASS
TC-CLIN-MEANS-007	Allocation 2:1 on TC-002 inputs	n treat = 95, n control = 48	PASS
TC-CLIN-MEANS-008	Sensitivity table over SD ($\times 0.8 \dots 1.2$)	5 scenario rows; totals 82 @ SD=8, 182 @ SD=12	PASS
TC-CLIN-MEANS-009	Superiority without --delta	Exit $\neq 0$, 'requires --delta'	PASS
TC-CLIN-MEANS-010	Non-inferiority without --margin	Exit $\neq 0$, 'requires --margin'	PASS
TC-CLIN-MEANS-011	Equivalence with $ \delta \geq \text{margin}$	Exit $\neq 0$, ' $ \delta < \text{margin}$ '	PASS
TC-CLIN-MEANS-012	--power out of range (1.5)	Exit $\neq 0$, '--power' in output	PASS
TC-CLIN-MEANS-013	Unknown flag	Exit $\neq 0$, 'Unknown argument'	PASS
TC-CLIN-MEANS-014	Bypass protection — direct Rscript call fails	Exit $\neq 0$, 'RENV_PATHS_ROOT' in output	PASS

6.4 jrc_clinical_ss_props — 11/11 PASS

TC ID	Description	Expected	Result
TC-CLIN-PROPS-001	Superiority valid inputs $\rightarrow$ exit 0, report header	Exit 0, 'Clinical sample size' + 'binary endpoint'	PASS
TC-CLIN-PROPS-002	Superiority numeric (pc .70, pt .85, power .80, α .05/2)	n treat = 118, n control = 118, n TOTAL = 236	PASS
TC-CLIN-PROPS-003	Non-inferiority numeric, default p-treat (pc .85, M .10)	n/arm = 201, n TOTAL = 402, 'assumed equal to control'	PASS
TC-CLIN-PROPS-004	Equivalence TOST numeric (pc .80, M .15, power .80)	n TOTAL = 244, 'equivalence (TOST)' in output	PASS
TC-CLIN-PROPS-005	Superiority without --p-treat	Exit $\neq 0$, 'requires --p-treat'	PASS
TC-CLIN-PROPS-006	Superiority with p-treat = p-control	Exit $\neq 0$, 'different from p-control'	PASS
TC-CLIN-PROPS-007	Non-inferiority without --margin	Exit $\neq 0$, 'requires --margin'	PASS
TC-CLIN-PROPS-008	--p-control out of range (1.2)	Exit $\neq 0$, '--p-control' in output	PASS
TC-CLIN-PROPS-009	Sensitivity table over p-control ($\times 0.8 \dots 1.2$)	5 scenario rows (0.64...0.96); assumed row marked	PASS
TC-CLIN-PROPS-010	Dropout 0.10 inflation on TC-003 inputs	'ENROLL 224 + 224 = 448' in output	PASS
TC-CLIN-PROPS-011	Bypass protection — direct Rscript call fails	Exit $\neq 0$, 'RENV_PATHS_ROOT' in output	PASS

6.5 jrc_clinical_ss_survival — 11/11 PASS

TC ID	Description	Expected	Result
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TC-CLIN-SURV-001	Superiority valid inputs → exit 0, report header	Exit 0, 'Clinical sample size' + 'time-to-event endpoint'	PASS
TC-CLIN-SURV-002	Superiority numeric (HR .70, power .80, α .05/2, PE .6)	Events = 247, n/arm = 206, n TOTAL = 412	PASS
TC-CLIN-SURV-003	Non-inferiority numeric (HR 1.0, M 1.3, α .025/1, PE .5)	Events = 457, n TOTAL = 914	PASS
TC-CLIN-SURV-004	--framework equivalence rejected	Exit ≠ 0, 'not offered'	PASS
TC-CLIN-SURV-005	Superiority with HR = 1	Exit ≠ 0, 'different from 1'	PASS
TC-CLIN-SURV-006	NI margin ≤ 1	Exit ≠ 0, '--margin' in output	PASS
TC-CLIN-SURV-007	NI with HR ≥ margin	Exit ≠ 0, '--hr < --margin'	PASS
TC-CLIN-SURV-008	Missing --event-prob	Exit ≠ 0, '--event-prob' in output	PASS
TC-CLIN-SURV-009	Sensitivity table over P(event) (×0.8...1.2)	5 scenario rows (0.48...0.72); assumed row marked	PASS
TC-CLIN-SURV-010	Dropout 0.20 inflation on TC-002 inputs	'ENROLL 258 + 258 = 516' in output	PASS
TC-CLIN-SURV-011	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS

7. Deviations

No deviations were identified during OQ execution. All 36 test cases passed.

8. Requirements Traceability Matrix

User requirements UR-CLIN-001 ... UR-CLIN-036 map 1:1, in order, to test cases TC-CLIN-MEANS-001..014, TC-CLIN-PROPS-001..011, and TC-CLIN-SURV-001..011 (see JR-VP-CLIN-001 for the full matrix). Grouped by capability:

Requirement group	Capability	Test Cases
UR-CLIN-001..008	jrc_clinical_ss_means — frameworks, numerics, allocation, dropout, sensitivity	TC-CLIN-MEANS-001..008
UR-CLIN-009..013	jrc_clinical_ss_means — input validation	TC-CLIN-MEANS-009..013
UR-CLIN-014	jrc_clinical_ss_means — bypass protection	TC-CLIN-MEANS-014
UR-CLIN-015..018	jrc_clinical_ss_props — frameworks and numerics	TC-CLIN-PROPS-001..004
UR-CLIN-019..024	jrc_clinical_ss_props — input validation, sensitivity, dropout	TC-CLIN-PROPS-005..010
UR-CLIN-025	jrc_clinical_ss_props — bypass protection	TC-CLIN-PROPS-011
UR-CLIN-026..028	jrc_clinical_ss_survival — frameworks and numerics (events → subjects)	TC-CLIN-SURV-001..003
UR-CLIN-029..035	jrc_clinical_ss_survival — input validation, sensitivity, dropout	TC-CLIN-SURV-004..010
UR-CLIN-036	jrc_clinical_ss_survival — bypass protection	TC-CLIN-SURV-011

9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Clinical Study Design Module v1.0 is complete.

Result: PASS

All 36 automated test cases specified in JR-VP-CLIN-001 v1.0 were executed on 2026-07-14 and 36 passed. Results are derived from the OQ evidence file listed below.

- jrc_clinical_ss_means — two-arm sample size, continuous endpoint (superiority / non-inferiority / equivalence)
- jrc_clinical_ss_props — two-arm sample size, binary endpoint, risk difference (superiority / non-inferiority / equivalence)
- jrc_clinical_ss_survival — two-arm sample size, time-to-event endpoint, log-rank (superiority / non-inferiority)

Evidence of test execution is archived at: [clinical_oq_execution_20260714T214209.txt](#)

The Clinical Study Design module is released as version 1.0 of repos/clinical/. Any subsequent change to the scripts, wrappers, or help files within repos/clinical/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.